



Uncovering Gatsby Curves

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Abstract

Empirical findings suggest a positive correlation between inequality and social immobility, a phenomenon coined the Gatsby curve. This paper answers a simple question: When do Gatsby curves exist? We build a theoretical n -income environment in which parental investment and education improve the economic prospects of children. Gatsbian economies and Gatsby curves are formally defined, and we characterize the conditions under which they arise. A Gatsby curve arises when immobility (Trace of the transition matrix) and inequality (Gini coefficient) both change in the same direction following a change in an exogenous variable (an income level, the level of education). When an exogenous variable changes, the impact on inequality depends on a standard direct effect and on an indirect (composition) effect which accounts for the changes in the proportions of individuals with various incomes. We show that following an increase in some high income, immobility and inequality both increase if the indirect effect goes in the same direction as the direct effect. Thus, in such a case, the economy moves up a Gatsby curve. We also demonstrate that if the indirect effect goes in the same direction as the direct effect, and if education is a substitute to parental investment, then an increase in education leads to the economy moving down a Gatsby curve where it experiences both lower inequality and immobility. Finally, we show that an economy may go from being Gatsbian to non-Gatsbian, and *vice versa*.

Keywords Intergenerational mobility · Income inequality · Education

JEL Classification D31 · H52 · J31 · J62

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1 Introduction

In this paper, we provide an answer to a simple question: When do Gatsby curves exist?

Recent empirical work suggests a positive correlation between income inequality and social immobility, a phenomenon coined the “(Great) Gatsby curve” by Krueger (2012). Figure 1, taken from Corak (2013), depicts a positive correlation between income inequality and social immobility for 22 countries. Jantti et al. (2006) also find evidence of a Gatsby-style correlation for a small panel of anglo-saxon and Scandinavian countries.

A Gatsby curve has also been observed within countries such as Canada or the United States (Conolly et al., 2018, 2019). However, it is not universal. Fan et al. (2015) show evidence that China has a negative correlation between inter-generational social immobility and cross-sectional inequality among parents. Chetty et al. (2014) also cast doubt on the positive slope of the Gatsby curve for the United States, pointing out to measurements issues.

So far, the literature on Gatsby curves has been mostly empirical and has been largely agnostic on the mechanisms which may lead to their existence. Rightly so, most authors have been careful not to present the positive correlation between income inequality and social immobility as a causal one. Thus, despite the high quality of the empirical work that has brought to light the Gatsby curve relationship, a complete and satisfactory theoretical explanation of the phenomenon yet remains to be proposed. This led Benabou (2017) to write: “Thus, after lagging behind theory for some time, measurement has now moved substantially ahead, and a renewed effort at closing the gap is well overdue.”

Our paper is part of a literature attempting to fill this gap. It contributes to a theory of Gatsby curves in several ways.

First, we draw on the Becker and Tomes (1979) framework and build a model in which parental investment contributes to inequality and immobility. Parents earn some income and they invest in their children. Parental investment increases the likelihood that a child will earn a higher income when he becomes an adult. Thus, our framework encompasses mobility as

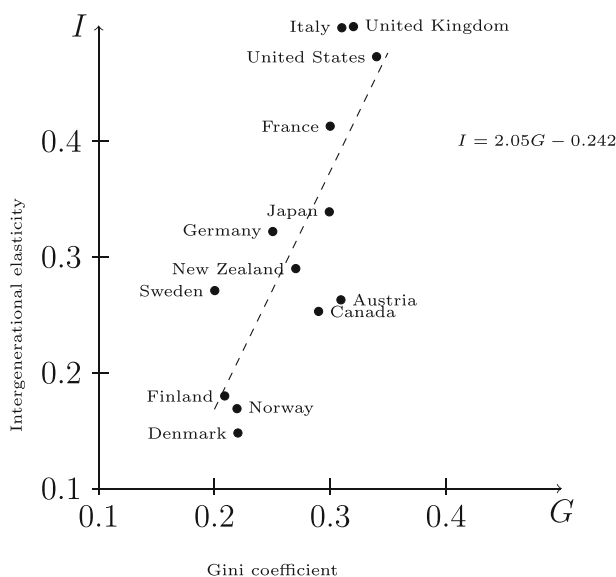


Fig. 1 The great Gatsby curve (Corak, 2013)

the probability that a child earns an income different from that of his parent. It further defines inequality using the proportions of individuals with various levels of income. Our framework is rich and tractable, providing closed form expressions for inequality, immobility, as well as a complete characterization of the steady-state transition matrix.

Second, we analyze the impact of an increase in education spending as well as that of an increase in the income of some individuals. We formally characterize the conditions under which an economy may experience a simultaneous increase in inequality (Gini coefficient) and immobility (Trace of the transition matrix) after such changes. An economy is then said to be x -Gatsbian when a change in x leads to a simultaneous increase (or a simultaneous decrease) in inequality and immobility. This is equivalent to moving along the positively-sloped portion of the x -expansion path in the inequality-immobility space. We characterize the conditions under which the economy is x -Gatsbian so that a Gatsby curve arises.

Third, we show that when an exogenous variable changes, the impact on inequality depends on a standard direct effect and on an indirect (composition) effect which accounts for the changes in the proportions of individuals with various incomes.

As for those changes in proportions, we show algebraically that inequality is a non-monotonic function of the proportion of individuals with a given income in the economy. For example, starting from a situation in which everybody earns the lowest income, reducing their proportion (hence increasing the proportions of individuals with higher incomes) at first increases inequality. But when a proportion of individuals of a given income reaches a certain threshold, inequality starts — and keeps on — decreasing as this proportion keeps on increasing. Note that the non-monotonicity of inequality in our framework bears resemblance to the classic Kuznets-Lewis process in which the movement of workers from a low-paid rural sector to a higher paid urban sector initially leads to an increase in inequality, which eventually reaches a summit, and then declines.¹

Because of the non-monotonic behaviour of the Gini coefficient, changes usually viewed as inequality-increasing may instead reduce inequality. For example, as in the empirical findings in Barany and Siegel (2018), we show that a widening of available incomes may reduce inequality. This happens because the proportions of individuals with various incomes change endogenously. Thus, the indirect (composition) effect is key in determining the behaviour of inequality and, ultimately, the existence of Gatsby curves.

Among many results we obtain regarding the existence of Gatsby curves, we show that following an increase in some high income, immobility and inequality both increase if (a sufficient, non-necessary condition) the indirect effect on inequality goes in the same direction as the direct effect. Thus, in such a case, the economy moves up a Gatsby curve. We also show that if (a sufficient, non-necessary condition) the indirect effect on inequality goes in the same direction as the direct effect, and if education is a substitute to parental investment, then an increase in education leads to the economy moving down a Gatsby curve where it experiences both lower inequality and immobility. And we also show that if a Gatsby curve does not obtain following a change in the lowest income, then it must be the case that at least some composition effects on inequality go in a direction opposite to that of the direct effect.

Fourth and lastly, we show that an economy may transition from being Gatsbian to non-Gatsbian (and *vice versa*). In other words, the x -expansion path in the inequality-immobility space may have both positively and negatively-sloped portions.

¹ The Kuznets curve associated with such a process has been extensively studied and remains at the heart of a large literature. The classic papers in this literature are Kuznets (1955) and Lewis (1954). For a recent analysis of Kuznets curves (in the context of China), see Ravallion and Chen (2022).

The remainder of the paper is structured as follows. In Section 2, we offer a brief survey of the relevant literature. We then turn to the presentation of our general n -income economy in Section 3. There, we characterize parental investment, the transitional dynamics and the stationary state of the economy, and we provide our measures of social immobility and income inequality. In Section 3.5, we show why it is important to account for “composition effects” in assessing the impact of exogenous variables on inequality. In Section 4, to provide intuition, we momentarily leave our general n -income economy framework to analyze a restricted two-income economy. In Section 5, we return to our general n -income economy and we provide our main results. There, we formally define Gatsbian economies and Gatsby curves, and we provide a complete characterization of the conditions under which Gatsby curves arise, whether composition effects are present or not. In the following section, Section 6, we then examine situations in which a Gatsbian economy may transition to being non-Gatsbian (and *vice versa*). Concluding remarks follow.

2 Literature

To our knowledge, three theoretical papers have made key contributions to our understanding of the relationship between inequality and immobility: Solon (2004), Durlauf and Seshadri (2017), and Becker et al. (2018). In all of them, and as in the Becker and Tomes (1979) seminal paper, parents transmit their own advantage or disadvantage to their children. Thus, parents determine how mobile children are and how unequal the future distribution of income will be.

To be precise, in all those papers, an advantage in the form of a larger stock of human capital can be passed on to the next generation. In Solon (2004) and in Becker et al. (2018), the transmission occurs because parents invest in the human capital of their children. As is shown in these papers, when parents care for their children, wealthier parents — equivalent to those with a larger stock of human capital — invest more in their children than poorer parents. Thus, children of wealthier parents end up with a larger stock of human capital and that makes them wealthier. This is the approach we adopt. Alternatively, in Durlauf and Seshadri (2017)² and in Becker et al. (2018)³, a child inherits a level of human capital that is larger when that of his parent is larger. Because a larger stock of human capital makes an individual wealthier, children of wealthy individuals end up being wealthier themselves.

All the aforementioned papers also include public education as an important ingredient. For example, Solon (2004) obtains that income inequality and social immobility can be accentuated by education policies that favour the children of richer families, as compared to the universal provision of education which would presumably be consumed by children of all backgrounds. As for Becker et al. (2018), they show that the degree of complementarity of public education with parental investment plays a key role in determining whether mobility is improved or reduced by an increase in public education expenditures.

We introduce public education in our analysis and we also show that the degree of complementarity / substitutability between parental investment and public education plays an important role. As we will show, an increase in public education may generate a Gatsby

² Durlauf and Seshadri (2017) also introduce segregation and voting as mechanisms generating inequality and immobility.

³ Thus, in Becker et al. (2018), the transmission of advantages is achieved through both parental investment and inheritance by the child of the traits of his parents.

curve, but that is not guaranteed and that will depend on whether education is a complement or a substitute to parental investment.

While all these papers provide interesting answers as to the mechanisms that could be at play behind Gatsby curves, they nevertheless leave aside a number of important issues that we address in our analysis. Among them is the very nature of a Gatsby curve and that of the conditions under which they may be observed.

Also, our analysis allows us to explore considerations that have been ignored so far in the literature. For example, all the above papers model income as a continuous variable and they do not characterize a continuous equivalent of a probability transition matrix, describing how individuals transit from one income to another. **Our approach, in which incomes take discrete values, allows for a complete characterization of the transition matrix.** This is important because two identical stationary distributions of incomes may be associated with different patterns of inter-generational mobility. We can thus fully characterize our measures of inequality and immobility, incorporating transition dynamics, and therefore push further the analysis of Gatsby curves.

We find that inequality is a non-monotonic function of the proportion of individuals with a given income, and that because of composition effects, inequality may increase or decrease after an increase in the dispersion of incomes.⁴ Our model also makes clear that a Gatsby curve is a positively-sloped expansion path in the inequality-immobility space. Further, our modelling allows for the full characterization of the conditions under which a Gatsby curve will be observed as well as that under which an economy will transition from being Gatsbian to non-Gatsbian.

In the empirical literature, parental investment in children and education policies are often cited as potential explanations for the Gatsby curve (Duncan and Murnane, 2011; Black and Devereux, 2011). For example, Guryan et al. (2008) find a strong relationship between the level of education of parents, their incomes and the time they spend with their children.⁵ This is consistent with theoretical works that model parental investments in children and public education as distinct inputs into a child's human capital (Kaganovich and Zilcha, 1999; Benabou, 2017). Child-rearing practices may also play a role: because poorer parents are limited in their ability to mold their children's behaviour through pecuniary incentives, they may rely more on non-pecuniary incentives such as corporal punishment, which is harmful to children's development (Weinberg, 2001). Family resources and connections are also identified as a potential channel by both Corak (2013) and Becker and Tomes (1979). In particular, there appears to be a correlation between a father and his son's job (Corak and Piraino, 2011; Corak, 2013).

A number of papers have also explored the impact of public education on inequality and mobility.⁶ For example, Corak (2013) observes that higher returns to college education are associated with lower social mobility. Moreover, his data shows that wealthier families engage in more enrichment expenditures, such as "books, computers, high-quality child care, summer camps, private schooling, and other things that promote the capabilities of their children" (p.15). Mazumder (2015) uses cross-country data and find correlations between social immobility and the level of inequality in test scores measuring literacy and numeracy skills

⁴ In the existing literature, an increase in the dispersion of income necessarily leads to an increase in income inequality because the composition of the population is fixed.

⁵ We do not attempt here to survey this vast literature. Rather, we simply want to illustrate that the connection between parental investment and inequality and immobility has received attention from empiricists.

⁶ Again, we do not attempt to make justice to this important literature.

in children. Finally, Rauh (2017) obtains that an increase in public education expenditures may increase inequality.

Thus, the incorporation of parental investment and public education in models like ours, trying to explain Gatsby curves, is justified on empirical grounds.

3 General framework

We use an overlapping generation framework in which time is indexed by t . Each individual lives for two periods, has a single child in the second period of his life, and his sole decision is whether to invest in his child education.

At any time, the economy has n incomes ordered $w^1 < w^2 < \dots < w^n$, each with its associated proportion of parents $Z^{1,t}, Z^{2,t}, \dots, Z^{n,t}$, where $\sum_{i=1}^n Z^{i,t} = 1$. We refer to each type of parent as a w^k -parent, $k = 1 \dots n$. We denote the difference between two incomes by $\Delta_i^j \equiv w^j - w^i$. We use the terms “available incomes” and “income dispersion” to respectively refer to the w^i s and Δ_i^j s. Thus, what we call income dispersion does not account for the proportions of individuals with various income levels. Later, we will define and use an inequality measure (the Gini coefficient) which accounts for both income dispersions and the various proportions. Further, it is useful to denote average income by $\bar{w} \equiv w^1 + \sum_{j=2}^n \Delta_1^j Z^{j,t}$, a function of all proportions but $Z^{1,t}$.

We denote by e the exogenous level of public education. In what follows, education has an impact on the likelihood of obtaining higher incomes. We explore the case in which public education is a complement to parental investment, and that in which it is a substitute. When public education is a complement, it benefits more to children in which parents have invested, while when it is a substitute, children in which parents have not invested benefit more. As will later be shown, whether public education can generate Gatsby curves depends on this distinction.

3.1 Parental investment and public education

The decision of a w^k -parent amounts to either investing an amount $a > 0$ in his child ($\phi^k = a$), or to invest nothing at all ($\phi^k = 0$).⁷ The cost of parental investment for a w^k -parent is $(\kappa - cw^k)\phi^k$, where $\kappa > 0$ is a baseline cost and where $c > 0$ is an income specific cost reduction.⁸ Each parent draws c from a uniform distribution with support $[c_0, c_1]$, where

⁷ We assume that parental investment is discrete (i.e. 0 or some fixed positive amount $a > 0$) for convenience. This assumption is without consequences. In our earlier work (see Bouchard-St-Amant et al. (2020)), we assumed that parents could invest any positive amount in their child, and that the larger the investment, the larger the probability their child would earn a high income as an adult. This earlier analysis was performed for the two-income case and it lead to results that are identical to those obtained in the current general n -income economy with discrete parental investment.

⁸ Thus, we assume that the cost of investing in their children is (on average) lower for parents with a larger income (a larger w^k). Below, we also assume that utility is linear in income. Combined, these assumptions imply that individuals with a larger income invest more in their children than individuals with a lower income, thereby generating some social immobility. Further, these assumptions ensure that our analysis is tractable and in particular, that we can solve analytically for the steady state proportions in our n -income economy. Note that we could have obtained similar results (i.e. some social immobility) with the alternative assumptions that utility is strictly concave in consumption and that the cost of investing in their children is the same for all parents, but we would then have lost analytical tractability and, in particular, we would not have been able to solve for the steady state proportions.

Table 1 Transition probabilities from income w^k to income w^ℓ

Income w^ℓ	Transition probability
w^1	$1 - \sum_{j=2}^n \alpha^j p((\phi^k + 1)e)$
w^2	$\alpha^2 p((\phi^k + 1)e)$
w^3	$\alpha^3 p((\phi^k + 1)e)$
$\vdots$	$\vdots$
w^n	$\alpha^n p((\phi^k + 1)e)$

$c_1 - c_0 = 1$. A parent's type is thus fully identified by the pair (w^k, c) . For later use, we denote by y the set of exogenous cost parameters that are relevant for parental investment decisions: $y = \{\kappa, c_0, c_1\}$.

We analyze the mobility of children born to w^k —parents who will become w^ℓ —parents ($\ell = 1 \dots n$) with some probability. Our focus is on the relationship between parental investment, public education and mobility. We thus assume that the income transition probability from w^k to w^ℓ , $\ell > 1$, takes the reduced form $\alpha^\ell p((\phi^k + 1)e)$, where $1 \geq p(\cdot) \geq 0$ is a strictly increasing and concave function⁹ and $1 > \alpha^\ell > 0$ is a multiplier specific to income ℓ . The probability function $p(\cdot)$ can be viewed as a reduced form of the process by which the human capital of a child translates into a higher income in adulthood. In our analysis, we make explicit that this human capital is positively affected by parental investment and public education. Of course, other variables (not modelled here) could also affect the human capital of a child, for example the socio-economic characteristics of his family, or that of his neighbourhood, or more generally any network effect. We assume that $\sum_{j=2}^n \alpha^j p((\phi^k + 1)e) \leq 1$. Since probabilities must sum to one, the transition from w^k to w^1 is given by $1 - \sum_{j=2}^n \alpha^j p((\phi^k + 1)e)$. Transition probabilities are summarized in Table 1. For simplicity, we denote $p^a \equiv p((a + 1)e)$ and $p^0 \equiv p(e)$. Of course, $p^a - p^0 > 0$.

The expected income of a child who has benefited from parental investment ϕ^k can then be written as:

$$\begin{aligned} \mathbb{E}[w|\phi^k] &= \left(1 - \sum_{j=2}^n \alpha^j p((\phi^k + 1)e)\right) w^1 + \sum_{j=2}^n \alpha^j p((\phi^k + 1)e) w^j, \\ &= w^1 + \left(\sum_{j=2}^n \alpha^j \Delta_1^j\right) p((\phi^k + 1)e). \end{aligned} \quad (1)$$

Because $a > 0$, $p' > 0$, and $w^1 < w^j \forall j > 1$, it is immediately obtained that for $e > 0$, parental investment increases the expected income of a child: $(\mathbb{E}[w|a] > \mathbb{E}[w|0])$. Denote the difference in expected incomes by:

$$\mathbb{E}^\Delta \equiv \mathbb{E}[w|a] - \mathbb{E}[w|0] = \left(\sum_{j=2}^n \alpha^j \Delta_1^j\right) (p^a - p^0). \quad (2)$$

⁹ With discrete parental investment, the concavity of p is not required, but it turns out to be important when parental investment is continuous. See Bouchard-St-Amant et al. (2020) for details.

The impact of a change in income or in education on the difference in expected incomes $\mathbb{E}^\Delta$ is given by:

$$\begin{aligned}\frac{\partial \mathbb{E}^\Delta}{\partial w^j} &= \alpha^j (p^a - p^0) > 0 \quad \forall j > 1, & \frac{\partial \mathbb{E}^\Delta}{\partial w^1} &= - \sum_{j=2}^n \alpha^j (p^a - p^0) < 0, \\ \frac{\partial \mathbb{E}^\Delta}{\partial e} &= \sum_{j=2}^n \alpha^j \Delta_1^j [(a+1)p^{a'} - p^{0'}] \geq 0.\end{aligned}$$

Thus, the difference in expected incomes $\mathbb{E}^\Delta$ is increasing in w^j , $j > 1$, and decreasing in w^1 . As for an increase in education e , it has an ambiguous impact on $\mathbb{E}^\Delta$. The direction of the impact depends on the sign of $(a+1)p^{a'} - p^{0'}$. Because p is a strictly concave function, the marginal probability under positive parental investment a (i.e. $p^{a'} = p'((a+1)e)$) is smaller than that with an investment of 0 (i.e. $p^{0'} = p'(e)$). However, the impact of public education on p is magnified by a factor $(a+1)$ when a parent invests a (while this impact is only magnified by a factor of 1 when a parent invests 0). Those two forces determine the sign of $(a+1)p^{a'} - p^{0'}$. Intuitively, if $(a+1)p^{a'} - p^{0'} < 0$, then public education has a larger impact on children in which parents have invested 0 than on children in which parents have invested a . Public education can then be viewed as a substitute to parental investment. In the alternative case with $(a+1)p^{a'} - p^{0'} > 0$, public education can be viewed as complement to parental investment.

In what follows, we use the concepts of complementarity or substitutability of education (with respect to parental investment) when referring to $(a+1)p^{a'} - p^{0'}$:

Definition 1 (Substitutability and Complementarity of Education).

We refer to *education as a substitute* when $(a+1)p^{a'} - p^{0'} < 0$.

We refer to *education as a complement* when $(a+1)p^{a'} - p^{0'} > 0$.

We assume that all income, net of the cost of parental investment, is devoted to consumption and that parents maximize the sum of their consumption and of the expected income of their child.¹⁰

$$u(w^k, c, \phi^k) = w^k - (\kappa - cw^k)\phi^k + \mathbb{E}[w|\phi^k]$$

A parent thus solves the following problem:

$$\begin{aligned}\max_{\phi^k \in \{0, a\}} & w^k - (\kappa - cw^k)\phi^k + \mathbb{E}[w|\phi^k] \\ \Leftrightarrow \max & \{\mathbb{E}^\Delta - (\kappa - cw^k)a, 0\}\end{aligned}\tag{3}$$

The sign of the expression $\mathbb{E}^\Delta - (\kappa - cw^k)a$ guides a w^k -parent in his investment choice. Equaling this expression to zero defines a break-even point $\hat{c}^k$ given by:

$$\hat{c}^k = \frac{\kappa}{w^k} - \frac{\sum_{j=2}^n \alpha^j \Delta_1^j (p^a - p^0)}{w^k a}, \quad k = 1 \dots n.\tag{4}$$

¹⁰ The specific parental objective function we use entails incomplete altruism, but it provides us with tractability, as it allows us to later solve analytically for the transition matrix and the steady state proportions. If we assumed complete altruism, then parents would maximize their consumption (income minus parental investment) plus the discounted expected consumption of their child, i.e. $[w^k - (\kappa - cw^k)\phi^k] + \beta E[w - (\kappa - cw)\phi^k | \phi^k]$, where β is a discount factor. Such an assumption would certainly be consistent with complete altruism, but it would make our problem analytically intractable as today's investment would be a function of past and future parental investments.

Intuitively, a w^k -parent who draws a large enough cost reduction $c \geq \hat{c}^k$ will invest a in his child. However, if this same parent draws a relatively small cost reduction c (in the sense that $c < \hat{c}^k$), then he invests 0. Now, for $w^j > w^i$, it is immediately obtained that $\hat{c}^j < \hat{c}^i$, so the break-even cost reduction differs for parents of different types. In order to have interesting comparative statics, we assume that $c_0 < \hat{c}^k < c_1 \ \forall k > 1$. However, to simplify the exposition and without loss of generality, we assume $c_1 < \hat{c}^1$.¹¹ These assumptions imply that all w^1 -parents invest 0 ($\phi^{1*} = 0$), and that $\forall k > 1$, w^k -parents with $c \in [\hat{c}^k, c_1]$ invest a ($\phi^{k*} = a$). We provide a proof of this statement in Appendix A.1.

In Appendix A.2, we further show that $\partial \hat{c}^k / \partial w^j < 0 \ \forall j > 1$, and that $\partial \hat{c}^k / \partial w^1 > 0$. Thus, an increase in w^j , $j > 1$, leads to more (w^k, c) -parents investing a , while an increase in w^1 has the opposite effect. As for the sign of $\partial \hat{c}^k / \partial e$, it is ambiguous because the sign of $\partial \mathbb{E}^\Delta / \partial e$ is itself ambiguous. If education is a complement (and thus if $(a+1)p^{a'} - p^{0'} > 0$), then an increase in education favours children in which w^k -parents have invested a . It follows that an increase in education leads more w^k -parents to invest a , which is equivalent to saying that $\partial \hat{c}^k / \partial e < 0$. Alternatively, if education is a substitute, then $\partial \hat{c}^k / \partial e > 0$, which means that less w^k -parents invest a .

3.2 Transitory dynamics and immobility

The dynamics of the proportions $Z^{i,t}$, $i = 1 \dots n$ is described by the following Markov chain:

$$\begin{bmatrix} Z^{1,t+1} \\ Z^{2,t+1} \\ \vdots \\ Z^{n,t+1} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 - p^0 \sum_{j=2}^n \alpha^j & (\hat{c}^2 - \underline{c}) \left(1 - p^0 \sum_{j=2}^n \alpha^j\right) & (\hat{c}^n - \underline{c}) \left(1 - p^0 \sum_{j=2}^n \alpha^j\right) \\ & + (\bar{c} - \hat{c}^2) \left(1 - p^a \sum_{j=2}^n \alpha^j\right) & \dots & + (\bar{c} - \hat{c}^n) \left(1 - p^a \sum_{j=2}^n \alpha^j\right) \\ \alpha^2 p^0 & (\hat{c}^2 - \underline{c}) \alpha^2 p^0 + (\bar{c} - \hat{c}^2) \alpha^2 p^a & \dots & (\hat{c}^n - \underline{c}) \alpha^2 p^0 + (\bar{c} - \hat{c}^n) \alpha^2 p^a \\ \vdots & \vdots & & \vdots \\ \alpha^n p^0 & (\hat{c}^2 - \underline{c}) \alpha^n p^0 + (\bar{c} - \hat{c}^2) \alpha^n p^a & \dots & (\hat{c}^n - \underline{c}) \alpha^n p^0 + (\bar{c} - \hat{c}^n) \alpha^n p^a \end{bmatrix}}_{\equiv T} \begin{bmatrix} Z^{1,t} \\ Z^{2,t} \\ \vdots \\ Z^{n,t} \end{bmatrix}, \quad (5)$$

where T is the transition matrix.

As discussed in Dardanoni (1993), several measures of immobility exist (e.g. the second eigenvalue or the trace of the transition matrix). Below, we use the trace of T as our measure of immobility:

$$I \equiv \text{Tr}(T) = 1 + (p^a - p^0) \sum_{j=2}^n \alpha^j (\bar{c} - \hat{c}^j). \quad (6)$$

Note that, by construction, any measure of immobility based on T is independent of steady-state proportions. Thus, our measure of immobility does not depend on the economy being at the steady-state. We can establish the following:

Proposition 1 (Comparative Statics of Immobility).

1. $dI/dw^1 < 0$.
2. $dI/dw^k > 0$, $k > 1$.
3. $dI/de > 0$ if and only if education is complement.

¹¹ This assumption implies that all w^1 -parents invest 0. Alternatively, we could have assumed that some w^1 -parents invest a , while all w^n -parents invest a . Irrespective of the assumption, we obtain the same qualitative results in what follows.

4. $dI/de < 0$ if and only if education is a substitute.

Proof See Appendix A.3.

Proposition 1 states that an increase in w^1 reduces immobility I , while it is the contrary for an increase in w^k , $k > 1$.

To understand Proposition 1, first recall that to end up with a w^1 income is the worse possible outcome for an individual, while ending up with any income w^i , $i = 2, \dots, n$ is a better outcome. Second, recall that when a parent invests $a > 0$, the probability with which his child will earn any income w^i , $i = 2, \dots, n$ increases, while the probability with which his child will earn income w^1 goes down. From those first two points follows¹² that an increase in any income w^i , $i = 2, \dots, n$ translates into more parents investing $a > 0$ because the return from parental investment has increased. Conversely, an increase in income w^1 translates into less parents investing $a > 0$ because in such a case, the worse outcome is not as bad, implying that the return from parental investment decreases. It follows that an increase in w^1 leads to less w^k -parents investing a in their child and so to less w^k -parents transmitting their high income to their offspring. As a consequence, a larger proportion of children have an income different from that of their parent and immobility I is reduced. The intuition is reversed for an increase in w^k , $k > 1$. Also note that the impact of an increase in education again depends on its substitutability / complementarity. When education is a substitute, an increase in education leads to less w^k -parents investing a . Thus, less of them transmit their high income to their offspring which results in a reduction in immobility I . The opposite occurs when education is a complement.

3.3 Steady-state proportions and comparative statics

In this economy, a steady-state occurs when the proportions $Z^{i,t}$ no longer change over time. Let the steady-state proportions be denoted by Z^{i*} , $i = 1, \dots, n$. Using Eq. 5, we can obtain that steady-state proportions are as follows:

$$Z^{1*} = 1 - \frac{p^0 \sum_{j=2}^n \alpha^j}{2 - I}, \quad (7)$$

$$Z^{i*} = \alpha^i \frac{p^0}{2 - I}, \quad i = 2 \dots n. \quad (8)$$

We will see later that understanding changes in inequality requires an understanding of how steady-state proportions change with exogenous variables. Thus, the following proposition will be useful in what follows:

Proposition 2 (Comparative Statics of Steady-State Proportions).

1. $\partial Z^{1*} / \partial w^1 > 0$ and $\partial Z^{1*} / \partial w^j < 0$, $j > 1$,
2. $\partial Z^{i*} / \partial w^1 < 0$ and $\partial Z^{i*} / \partial w^j > 0$, $i > 1$, $j > 1$,
3. If education is a complement, then $\partial Z^{i*} / \partial e > 0$, $i > 1$.
4. If $\partial Z^{i*} / \partial e < 0$, $i > 1$, then education is a substitute.

Proof See Appendix A.4.

¹² See Appendix A.2.

Increasing income w^1 reduces the attractiveness of investing to transition to higher incomes, thereby reducing parental investments. As a result, the steady-state low-income proportion Z^{1*} increases. The converse is true with respect to increases in higher incomes ($w^j, j > 1$): the relative attractiveness of the lowest income is reduced and that of higher incomes is increased, making parental investment more attractive, thereby decreasing Z^{1*} . An equivalent interpretation holds for the proportions of higher incomes $Z^{i*}, i > 1$: increasing the lowest income w^1 lowers those proportions, while increasing higher incomes ($w^j, j > 1$) does the opposite.

3.4 Inequality

We use the Gini coefficient G to account for income inequality in the economy. We focus first on its characterization for any proportions $Z^i, i = 1, \dots, n$. Later, we will analyze G when proportions are those of the steady-state.

Using $Z^1 = 1 - \sum_{i=2}^n Z^i$, we use a representation of the Gini coefficient which does not depend on Z^1 . Indeed, the Gini coefficient can be shown to satisfy (see Appendix A.5):

$$G\bar{w} = \sum_{i=2}^n \Delta_1^i Z^i (1 - Z^i) - 2 \sum_{i=3}^n \sum_{j=2}^{i-1} \Delta_1^j Z^j Z^i. \quad (9)$$

Function G is globally concave and non-negative for all Z^j such that $\sum_{j=2}^n Z^j < 1$.¹³ It has a global maximum, which is the solution to a set of $n - 1$ partial derivatives (i.e. first order conditions) of G given by (see Appendix A.8):

$$\frac{\partial G}{\partial Z^k} \bar{w} = \Delta_1^k \left(1 - G - 2 \sum_{j=k}^n Z^j - 2 \sum_{j=2}^{k-1} \frac{\Delta_1^j}{\Delta_1^k} Z^j \right), \quad k = 2, \dots, n, \quad (10)$$

where the last term within the parentheses must be ignored when $k = 2$.¹⁴ The $n - 1$ partial derivatives of Eq. 10 can jointly equal zero at only one point, that is when:

$$\begin{aligned} Z^n &= \hat{Z}_1^n = \frac{\sqrt{w^1}}{\sqrt{w^1} + \sqrt{w^n}} \in]0, 1/2[, \\ Z^1 &= 1 - \hat{Z}_1^n \in]1/2, 1[, \quad Z^j = 0 \quad \forall j = 2, \dots, n - 1. \end{aligned} \quad (11)$$

At this global maximum, G attains a value (see Appendix A.9):

$$G = 2\hat{Z}_1^n - 1 = \frac{\sqrt{w^n} - \sqrt{w^1}}{\sqrt{w^n} + \sqrt{w^1}}. \quad (12)$$

Solving individually each partial derivative of Eq. 10 yields $n(n - 1)/2 - 1$ constrained maxima, each characterized by $\hat{Z}_j^i \equiv \sqrt{w^j}/(\sqrt{w^i} + \sqrt{w^j}), j < i$, and some other proportion j filling the remainder of the population ($Z^j = 1 - \hat{Z}_j^i$). Those can be understood as the inequality maximizing proportions if they were the sole incomes in the economy. In that case, inequality is given by $G = (\sqrt{w^i} - \sqrt{w^j})/(\sqrt{w^i} + \sqrt{w^j})$. Those constrained maxima are well ordered: a constrained maximum $\hat{Z}_j^i$ yields a higher level of inequality than another

¹³ There are n points such that $G = 0$, which are the points where one proportion Z^i equals one. All other points yield a strictly positive value of inequality. See Appendices A.6 and A.7.

¹⁴ We use the convention that $\sum_{j=n}^n Z^j = Z^n$.

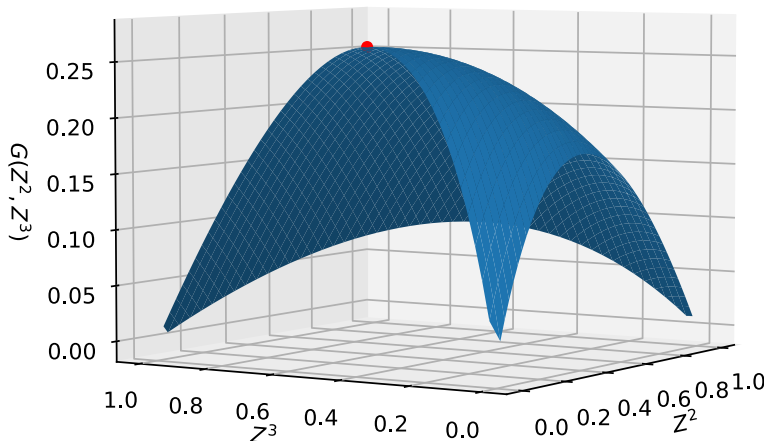


Fig. 2 Inequality G in a three-income economy with $w^1 = 1$, $w^2 = 2$, and $w^3 = 3$. The global maximum is shown in red

$\hat{Z}_{j'}^{i'}$ if and only if $w^i/w^j > w^{i'}/w^{j'}$. Since w^n/w^1 is the highest possible ratio, $\hat{Z}_1^n$ is the global (unconstrained) maximum.¹⁵ Figure 2 depicts inequality G in the (Z^2, Z^3) -space for the case of a three-income economy. The global maximum is shown in red, while the constrained maximum is the highest point along the Z^2 axis.

A careful inspection of Eq. 10 reveals that the partial derivatives are ordered for any proportions in the economy. In Eq. 10, decreasing k simply shifts one proportion from the middle sum to the last sum. The middle sum has a coefficient of -2 , which is smaller than the coefficient $-2\Delta_1^j/\Delta_1^k$ of the last sum. Hence, increasing k necessarily increases the term between brackets. As the term Δ_1^k outside the brackets also increases with k , the following proposition obtains:

Proposition 3 (Ordering of Partial Derivatives).

For any proportion Z^i , $i > 1$, such that $i < j$, then:

$$\frac{\partial G}{\partial Z^i} < \frac{\partial G}{\partial Z^j}.$$

In particular, if the partial derivative with respect to Z^i , $i > 1$, is positive, then all partial derivatives with respect to Z^j , $j > i$, are also positive.

Proof See Appendix A.8.

This observation provides the basis of a definition which turns out to be useful when we later characterize the circumstances under which Gatsby curves arise:

Definition 2 (w^i -Dominated Economy)

An economy is **w^i -dominated** if, for $i > 1$:

$$1 - G - 2 \sum_{j=i}^n Z^j - 2 \sum_{j=2}^{i-1} \frac{\Delta_1^j}{\Delta_1^i} Z^j < 0. \quad (13)$$

¹⁵ As a corollary, an economy cannot exhibit maximal inequality if the proportions of individuals with incomes different from w^1 and w^n are non-zero.

Intuitively, each partial derivative in Eq. 10 can be viewed as generating an iso-curve when it is set equal to zero (i.e. when $\partial G / \partial Z^i = 0$). An economy is then w^i -dominated at given proportions $(Z^1, \dots, Z^n)$ if, at those proportions, $\partial G / \partial Z^i < 0$.

From Proposition 3, because partial derivatives are ordered, we obtain the following corollary :

Corollary 1 For $i > 2$, a w^i -dominated economy is also a w^{i-1} -dominated economy.

In Fig. 3, we represent, for a three-income economy (w^1, w^2, w^3) , the w^i -dominated regions in the (Z^2, Z^3) space.

3.5 Composition effects and changes in inequality

A fundamental point that seems to have gone unnoticed in the literature on Gatsby curves is that the Gini coefficient is non-monotonic, in the sense that an increase in *any* available income may lead to either an increase or a decrease in inequality at some vector of proportions Z^k , $k > 1$, depending on whether or not, and for which i , the economy is w^i -dominated. For example, in Fig. 3, G is depicted as a function of Z^2 and Z^3 , with its global maximum reached at $\hat{Z}_1^3$. Inequality increases with Z^3 only when the economy is not w^3 -dominated (below iso-curve $\partial G / \partial Z^3 = 0$), while it decreases with Z^3 when the economy is w^3 -dominated (above iso-curve $\partial G / \partial Z^3 = 0$).

This implies that a change in inequality that is caused by a change in an exogenous variable should be decomposed as the sum of a direct effect and of an indirect effect which accounts for changes in steady-state proportions Z^{i*} , $i = 1, \dots, n$. Changes in steady-state proportions are referred to as *composition effects* and they determine the sign and magnitude

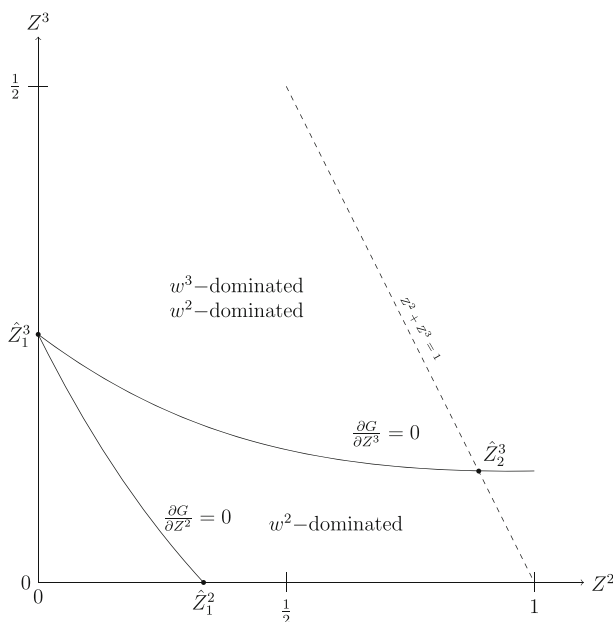


Fig. 3 Proportions iso-curves ($\partial G / \partial Z^i = 0$, $i = 2, 3$) and w^i -dominated regions ($i = 2, 3$) in a three-income economy

of the indirect effect. As we will see, if the indirect effect has a sign opposite to that of the direct effect and is of greater magnitude, then counter intuitive results are obtained. For instance, an increase in the highest available income w^n can lead to a decrease in inequality.

Below, in Section 3.5.2, we evaluate how inequality changes when accounting for both direct and indirect effects. Taking into account the indirect effect (hence composition effects) is certainly the most comprehensive approach. However, for two reasons, we first carry a constrained analysis in which composition effects are absent. First, the absence of composition effects can be understood as an analysis of the immediate (short run) impact of an exogenous variable, i.e. prior to reaching new steady-state proportions. Second, it allows us to show that the conditions under which Gatsby curves arise are not the same with or without composition effects. This distinction could prove helpful in empirical analyses, as those analyses could yield opposite results when focusing on different time horizons (hence allowing or not for composition effects).

3.5.1 Direct effects

The direct effect of income w^k on inequality G is characterized by the following partial derivative (see Appendix A.10):

$$\frac{\partial G}{\partial w^k} = \frac{Z^k}{\bar{w}} \left[\sum_{i=1}^{k-1} Z^i - \sum_{i=k+1}^n Z^i - G \right], \quad (14)$$

where the sums with inverted indices must be ignored when $k = 1$ or $k = n$. Again, the structure of the direct effects shows a clear ordering :

Proposition 4 (Ordering of Direct Effects of Incomes on Inequality)

1. $\partial G / \partial w^1 < 0$
2. $\partial G / \partial w^{k+1} > \partial G / \partial w^k$,
3. $\partial G / \partial w^n > 0$

Proof See Appendix A.10. □

Combined, these results imply that there exists a unique income w^ℓ such that all direct effects with respect to this income, or an higher income, are positive ($\partial G / \partial w^i \geq 0$ for $i = \ell, \dots, n$), and all direct effects with respect to lower incomes are negative ($\partial G / \partial w^i < 0$ for $i = 2, \dots, \ell - 1$).

For completeness, it should be noted that because education e has no impact on available incomes (no impact on w^i , $i = 1, \dots, n$), it has no direct effect on inequality ($\partial G / \partial e = 0$) for given proportions. However, education has an impact on the proportions of individuals with various available income levels (that is, an impact on Z^i , $i = 1, \dots, n$) so it will impact inequality through the indirect (composition) effect.

3.5.2 Indirect effects

It turns out to be useful to re-write Eq. 10 as follows:

$$\frac{\partial G}{\partial Z^k} = \Delta_1^k \left(\sum_{i=1}^{k-1} Z^j - \sum_{j=k+1}^n Z^j - G - Z^k - 2 \sum_{j=2}^{k-1} \frac{\Delta_1^j}{\Delta_1^k} Z^j \right),$$

$$= \Delta_1^k \left(\frac{\bar{w}}{Z^k} \frac{\partial G}{\partial w^k} - Z^k - 2 \sum_{j=2}^{k-1} \frac{\Delta_1^j}{\Delta_1^k} Z^j \right). \quad (15)$$

Hence, there is a key connection between the direct effects of incomes, given by Eq. 14, and the $n - 1$ partial derivatives of G with respect to proportions (given in Eq. 10 and now Eq. 15) that we used to define w^i -dominated economies. This leads to the following proposition:

Proposition 5 (Direct Effects and w^i -Dominated Economies) *If $\partial G / \partial w^k < 0$, then the economy is w^i -dominated, for some $i \geq k$. Conversely, if an economy is not w^k -dominated, then $\partial G / \partial w^k > 0$.*

Proof A careful use of Definition 2 constitutes the proof of this proposition. If $\partial G / \partial Z^k < 0$, then Eq. 15 is necessarily negative, which implies that there is an $i \geq k$ for which the economy is w^i -dominated. The second part of the proposition is obtained in a similar fashion.

Using the expressions found for steady-state proportions Z^{i*} , we can now incorporate the composition effects in the complete expression characterizing the impact of a change in a variable $x \in \{e, w^1, \dots, w^n\}$ on inequality G :

$$\frac{dG}{dx} \bar{w} = \sum_{i=2}^n \underbrace{\left\{ \Delta_1^i \left[\underbrace{1 - G - 2 \sum_{j=i}^n Z^j - 2 \sum_{j=2}^{i-1} \frac{\Delta_1^j}{\Delta_1^i} Z^j}_{w^i\text{-dominated } (-) \text{ or non } w^i\text{-dominated } (+)} \right] \frac{\partial Z^{i*}}{\partial x} \right\}}_{\text{Indirect effect}} + \underbrace{\frac{\partial G}{\partial x}}_{\text{Direct effect}}. \quad (16)$$

The signs of the various bracketed terms indicate whether the economy is w^i -dominated. These signs, of course combined with the magnitude of the term, determine the indirect effect on G through the change in the proportions Z^{i*} , $i = 2, \dots, n$. This leads to the following proposition:

Proposition 6 (Changes in Inequality with Composition Effects)

1. *If the economy is not w^k -dominated for all $k > 1$, then $dG/dw^1 < 0$ and $dG/dw^j > 0$, $j > 1$.*
2. *If $dG/dw^1 > 0$ or $dG/dw^j < 0$, $j > 1$, then there exists an ℓ such that the economy is w^ℓ -dominated.*
3. *If the economy is not w^k -dominated for all $k > 1$, and education is a complement (substitute), then $dG/de > 0$ ($dG/de < 0$).*
4. *If education is a complement (substitute) and $dG/de < 0$ ($dG/de > 0$), then there exists an ℓ such that the economy is w^ℓ -dominated.*

Proof See Appendix A.11.

A key element of Proposition 6 is what it does *not* contain. If an economy is w^k -dominated for some $k > 1$, then inequality G can move in any direction after a change in income, depending on the indirect effect. This includes most notably the possibility of a decrease in inequality G following and increase in a higher available income w^j , $j > 1$. It should further be noted that because a w^i -dominated economy is also a w^{i-1} -dominated economy, such a possibility is more likely for income w^ℓ than for income w^k if $\ell > k > 1$.

4 Building-up intuition: a two-income economy

In this section, we momentarily leave our general setup to focus on a simpler two-income economy. Such a simplified setup is useful to build intuition for most of the results we later provide when we return to our broader n -income setup. Further, we will also use this simple two-income framework when we present, in Section 6, the possibility for an economy to transition from being Gatsbian to non-Gatsbian.

We thus consider a two-income economy with $w^2 > w^1$ and we let $\alpha^2 = 1$. In such a case, the main equations of the steady-state economy boil down to:

$$\hat{c}^2 = \frac{\kappa}{w^2} - \frac{\Delta_1^2}{w^2} \frac{p^a - p^0}{a}, \quad \bar{w} = w^1 + \Delta_1^2 Z^{2*} \quad (17)$$

$$Z^{1*} = 1 - Z^{2*}, \quad Z^{2*} = \frac{p^0}{2 - I}, \quad (18)$$

$$I = 1 + (c_1 - \hat{c}^2)(p^a - p^0), \quad G\bar{w} = \Delta_1^2 Z^{2*}(1 - Z^{2*}). \quad (19)$$

4.1 Immobility

One can verify that an increase in w^1 reduces immobility I ($dI/dw^1 = -(p^a - p^0)^2/(aw^2) < 0$), while it is the contrary for an increase in w^2 ($dI/dw^2 = (\hat{c}^2/w^2 + (p^a - p^0)/w^2)(p^a - p^0) > 0$). Intuitively, an increase in w^1 leads to less w^2 -parents investing a in their child and so to less w^2 -parents transmitting their high income to their offspring. As a consequence, a larger proportion of children have an income different from that of their parent and immobility I is reduced. The intuition is reversed for an increase in w^2 .

The impact of an increase in education depends on its substitutability / complementarity ($dI/de = (c_1 - \hat{c}^2)((a + 1)p^{a'} - p^0)$). When education is a substitute, an increase in education leads to less w^2 -parents investing a . Thus, less of them transmit their high income to their offspring which results in a reduction in immobility I . The opposite occurs when education is a complement.

4.2 Steady-state proportions

Steady-state proportion Z^{2*} varies as immobility I does ($\partial Z^{2*}/\partial x = p^0/(2 - I)^2 \partial I/\partial x$ for $x \in \{e, w^1, w^2\}$). For example, when an exogenous variable increases the incentive to invest in a child's education (e.g. an increase in w^2 or an increase in e when education is a complement), more w^2 -parents invest in their child (hence transmitting their high income) so that both immobility and the steady-state proportion of w^2 -parents increase.

4.3 Composition effects

Absent composition effects, inequality G is increasing in w^2 ($\partial G/\partial w^2 \bar{w} = Z^{2*}(1 - Z^{2*} - G) > 0$) and decreasing in w^1 ($\partial G/\partial w^1 \bar{w} = -(Z^{2*} + G)(1 - Z^{2*}) < 0$).¹⁶ Thus, without composition effects, an increase in income dispersion leads to an increase in inequality.

However, allowing for composition effects, thereby accounting for the impact of income dispersion on steady-state proportions Z^{1*} and Z^{2*} , may reverse the latter result.

¹⁶ Note that without composition effects, education e has no impact on inequality G .

Table 2 Changes in G without and with composition effects

Situation	w^1	w^2	Z^{1*}	Z^{2*}	σ^2	G
Initial distribution	1	4	40%	60%	2.16	0.26
$\uparrow w^2$	1	5	40%	60%	3.84	0.28
$\uparrow w^2$ & Composition effects	1	5	30%	70%	3.36	0.22

In Table 2, we provide an illustrative case in which increasing w^2 (hence the dispersion of available incomes) would result in more inequality absent composition effects, but in less inequality in the presence of composition effects. Starting from the initial distribution with $w^1 = 1$, $w^2 = 4$ (the first line), and absent compositions effect so that the proportions of w^1 - and w^2 -parents remain fixed, an increase in w^2 from 4 to 5 leads to an increase in the variance (column σ^2), from 2.16 to 3.84, and in the Gini coefficient, from 0.26 to 0.28 (second line). But, accounting for composition effects, the same increase in w^2 from 4 to 5 has a different impact, increasing variance while *lowering* inequality (third line).¹⁷ Thus, an increase in the dispersion of incomes may translate into less inequality when one accounts for composition effects.

Such a possibility is related to the concept of a w^2 -dominated economy. In a two-income setup, inequality is maximized when the following partial derivative is set to zero:

$$\frac{\partial G}{\partial Z^2} \bar{w} = \Delta_1^2 (1 - G - 2Z^2).$$

This equation then yields the single maximum $\hat{Z}_1^2 = \sqrt{w^1}/(\sqrt{w^1} + \sqrt{w^2})$. Following our earlier definition, the economy is w^2 -dominated when this partial derivative is negative. In such a w^2 -dominated economy (i.e. when $Z^{2*} > \hat{Z}_1^2$), an increase in Z^{2*} leads to a *reduction* in inequality G (see Fig. 4). As was mentioned in the Introduction, the latter phenomenon bears resemblance to the to the classic Kuznets-Lewis process in which the movement of workers from a low-paid rural sector to a higher paid urban sector leads to inequality behaving as in Fig. 4. Thus, one could view the graph depicted in Fig. 4 as a Kuznets curve.

This simple two-income analysis provides a glimpse of what is to come. In determining the joint behaviour of immobility and inequality, whether and for what incomes an economy is w^i -dominated will be key.

5 Gatsbian economies

We now return to our n -income economy. In the literature, a Gatsby curve is a graph depicting a positive relationship between income inequality and social immobility (see Fig. 5). We use this relationship as the starting point of our formal definition of Gatsby curves and characterize the circumstances in which they occur. An economy will be said to be Gatsbian when social immobility and income inequality move in tandem, meaning that both increase (or decrease) after a change in an exogenous variable. When an exogenous parameter (e.g. income levels, education) changes, the economy moves towards a new stationary state at which inequality G and immobility I take some new values. Therefore, one would observe what the literature

¹⁷ Here, it is simply assumed that the proportion of w^2 -parents increases (exogenously in this example — in our analysis, those changes are of course endogenous) from 60% to 70%.

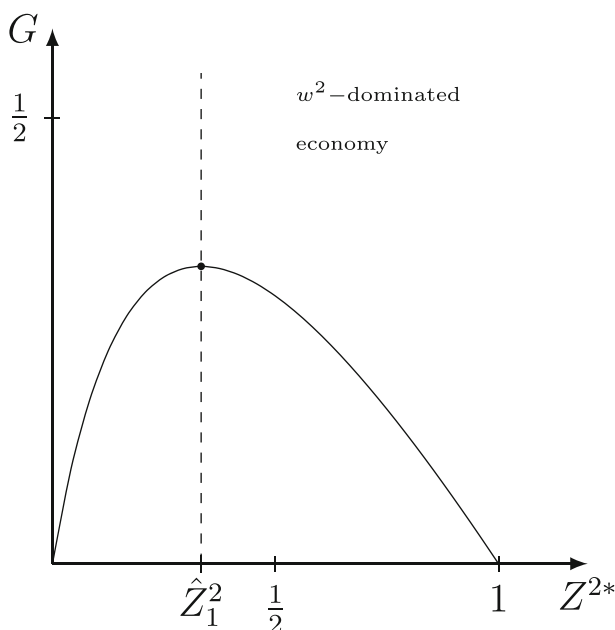


Fig. 4 Inequality as a function of Z^{2*} . G is maximized at $\hat{Z}_1^2$

has called a Gatsby curve when G and I move in the same direction following a change in a parameter $x \in \{e, w^1, \dots, w^n\}$. As the last sentence suggests, the relationship is not a causal one.

Definition 3 (Gatsbian Economy).

For $x \in \{e, w^1, \dots, w^n\}$, an x -**expansion path** is the curve of stationary state points $(G(x), I(x))$ that is induced by the values of x . We say that an economy is **Gatsbian** with respect to a variable x , or is equivalently **x -Gatsbian** if, at its stationary state, the economy is on the positively-sloped segment of the x -expansion path:

$$x\text{-Gatsbian} \Leftrightarrow \frac{dI(x)/dx}{dG(x)/dx} > 0. \quad (20)$$

Otherwise, the economy is non-Gatsbian.

Definition 4 (Gatsby Curve).

A **Gatsby curve** is a positively-sloped segment of an expansion path in the (G, I) space.

From these definitions, it should be clear that using the signs of the total derivatives of immobility and inequality is useful to characterize the cases under which an economy is x -Gatsbian. Table 3 classifies the possible cases, with two of them entailing a Gatsbian economy.¹⁸

Intuitively, the top-left case of Gatsbian economies is that in which a change in x moves the economy up along a Gatsby curve. The bottom-right case is that in which the economy moves down along a Gatsby curve after a change in x .

¹⁸ As they bear no interest, we omit the cases in which total derivatives equal zero.

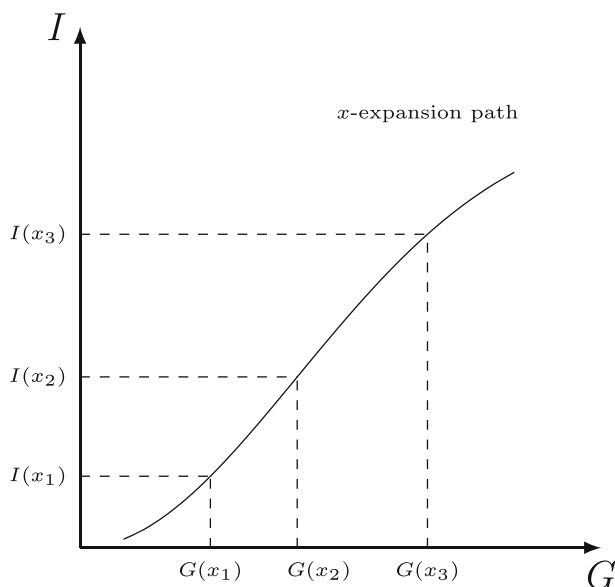


Fig. 5 A positively-sloped x -expansion path is a Gatsby curve

With our earlier results on the impact of variables $x \in \{e, w^1, \dots, w^n\}$ on inequality G (Propositions 4 and 6) and immunity I (Proposition 1), we can now identify the conditions under which a Gatsbian economy and a Gatsby curve will obtain. We distinguish between the cases in which composition effects are absent from those in which they take place. We begin with the case in which composition effects are assumed away. We can obtain the following:

Proposition 7 (Gatsbian Economies *without* Composition Effects).

1. Without composition effects, the economy is not e -Gatsbian.
2. Without composition effects, the economy is w^1 -Gatsbian. As w^1 increases, the economy moves down along a Gatsby curve.
3. Without composition effects, the economy is w^n -Gatsbian. As w^n increases, the economy moves up along a Gatsby curve.
4. Without composition effects, if the economy is w^k -Gatsbian, $k > 1$, then it is also w^{k+1} -Gatsbian. As w^k increases, the economy moves up along a Gatsby curve.
5. Without composition effects, if the economy is not w^k -Gatsbian, then it is not w^{k-1} -Gatsbian for $k - 1 > 1$.

Proof Corollaries of Propositions 1 and 4.

Table 3 Classification of Gatsbian and non-Gatsbian economies

		Inequality	
		$dG/dx > 0$	$dG/dx < 0$
Immobility	$dI/dx > 0$	x -Gatsbian	Non-Gatsbian
	$dI/dx < 0$	Non-Gatsbian	x -Gatsbian

Thus, without composition effects, an increase in income dispersion (e.g. a decrease in w^1 and/or an increase in some higher income w^k , $k > 1$) necessarily leads to a movement up along a Gatsby curve in which both inequality and immobility increase.

Let us now turn to the cases in which there are composition effects.

Proposition 8 (Gatsbian Economies with Composition Effects).

1. If an economy is not w^i -dominated for all $i > 1$, then it is a w^k -Gatsbian economy for all $k > 1$, and it is not a w^1 -Gatsbian economy. In such a case, an increase in w^k , $k > 1$, moves the economy up along a Gatsby curve.
2. If an economy is w^k -Gatsbian for $k > 1$, then it is w^{k+1} -Gatsbian.
3. If an economy is not w^1 -Gatsbian, then there is at least one $\ell > 1$ such that it is w^ℓ -dominated.
4. An economy may be non-Gatsbian with respect to w^1 or w^k , $k = 2, \dots, n$.
5. If an economy is not w^i -dominated for all $i > 1$ and education is a complement (substitute), then it is e -Gatsbian. In such a case, increasing e moves the economy up (down) along a Gatsby curve.
6. If an economy is not e -Gatsbian and education is a complement (substitute), then there is at least one $\ell > 1$ such that it is w^ℓ -dominated.

Proof 1. First note that immobility I increases with w^k ($k > 1$). Second, recall from Proposition 5 that if an economy is not w^i -dominated for all $i > 1$, then the direct effect $\partial G/\partial w^k > 0$ for all $k > 1$. Further, since the composition effect $\partial Z^{j*}/\partial w^k > 0$ for all $j > 1$ and for all $k > 1$, the indirect effect in Eq. 16 is necessarily positive. It follows that dG/dw^k is necessarily positive for all $k > 1$, proving the result.

2. This is a direct consequence of the ordering of partial derivatives (see Proposition 3 and Corollary 1).
3. Recall that immobility I decreases with w^1 . Hence, if an economy is not w^1 -Gatsbian, then dG/dw^1 is necessarily positive. Recall from Proposition 4 that the direct effect of w^1 on G is negative: $\partial G/\partial w^1 < 0$. Thus, for $dG/dw^1 > 0$, it must be that the indirect effect in Eq. 16 is positive. Recall from Proposition 2 that $\partial Z^{j*}/\partial w^1 < 0$, for $j = 2, \dots, n$. Thus, for the indirect effect to be positive it must be that there is at least one $\ell > 1$ such that it is w^ℓ -dominated.
4. Consider an economy where $Z^{n*} \approx 1$. In such case, it is not w^k -gatsbian for any k .
5. If an economy is not w^i -dominated for all $i > 1$ and education is a complement, then dG/de is necessarily positive. If education is a complement, then $dI/de > 0$, from which the result follows.
6. If the economy is not e -gatsbian and education is a complement so that $dI/de > 0$, then it must be that $dG/de < 0$. Since $\partial Z^{i*}/\partial e > 0$, it follows that some partial derivatives with respect to Z^{i*} in Eq. 16 are necessarily negative for some i (w^i -dominated) so that the indirect effect is negative. The case where education is a substitute is shown in a similar fashion.

It should be noted that there is no guarantee that the existence of a Gatsby curve in the presence of composition effects implies the existence of a Gatsby curve when there are none. The converse is also true. Also note that Gatsby curves are more likely to emerge when the economy is not w^k -dominated for all k , $k > 1$. Indeed, Proposition 8 shows that if an economy is not w^k -dominated for all k , then an increase in income dispersion (i.e. an increase in w^n and/or a decrease in w^1) translates into a movement up along a Gatsby curve.

However, in other cases, it is possible for an increase in income dispersion to translate into a reduction in inequality. In such cases, the economy must be w^k -dominated for at least some k and the conventional intuition may be reversed. Also, if the economy is not w^k -dominated for all k , then an increase in some income w^k , $k > 1$, increases immobility, but inequality may be decreasing. Such a case would then simply yield no Gatsby curve at all.

A last point to note is that in Propositions 7 and 8, Gatsbian economies obtain when the economy is not w^k -dominated for all k . But, of course, since some of those conditions are non necessary, it remains possible for Gatsby curves to occur when the economy w^k -dominated for some k .

In our analysis, we have used the Gini index as our measure of inequality, and the Trace of the transition matrix as our measure of social immobility. How do our results generalize to the use of other measures of inequality and immobility?

Start with inequality. As was shown, depending on the sign of the derivative of the inequality measure with respect to available incomes, an economy can either be Gatsbian or Non-Gatsbian due to the indirect (composition) effect. Thus, an inequality measure that allows for a change in the sign of this derivative implies that an economy can be both Gatsbian or Non-Gatsbian. This, it turns out, is true of any measure of inequality that is concave in incomes. By definition, concave (and differentiable) functions have increasing and decreasing derivatives, depending on the side of the extrema on which the analysis is performed. Therefore, concave measures of inequality such as the Theil index or the variance of log earnings should yield results similar to those obtained with the Gini index.

Consider now social immobility. Our measure, the Trace of the transition matrix, is monotonic in incomes (see Appendix A.3). Thus, all our results on the existence or not of a Gatsbian economy are mainly driven by the changing behaviour of our inequality measure. Of course, if our immobility measure was not monotonic in incomes (as it would be with an alternative measure like the second eigenvalue of the transition matrix), then there would be even more scenarios under which the economy would be Gatsbian (inequality and immobility affected in the same manner by a change in an exogenous variable) or not (inequality and immobility affected in a different manner by a change in an exogenous variable).

To summarize, the combination of the concavity of the Gini index and of the monotonicity of the Trace is a sufficient (non-necessary) condition for our results to obtain. But these results could clearly be obtained using other measures, although the specific conditions under which that would be the case would obviously be different.

6 From Gatsbian to Non-Gatsbian (and Vice Versa)

In this section, we show that an economy can be e -Gatsbian below a certain level of education, and then become non-Gatsbian when the level of education crosses a threshold. The reverse is also possible. We go back to a two-income economy (with the same assumptions as in Section 4) and present two examples to illustrate those possible transitions.

The transitions can be understood intuitively. Consider the case of a low-income dominated e -Gatsbian economy when education is a complement (see Proposition 8). As was shown above, in such an economy, immobility increases with education. Now suppose that in this e -Gatsbian economy, everybody is low-income (thus $Z^{1*} = 1$). Then, replacing a low-income individual by a high-income one increases inequality. From above, we know that as the level of education increases, proportion Z^{1*} diminishes (and Z^{2*} increases), and this translates into an increase in inequality. Thus, initially, both immobility and inequality increase with

education. But, as we show below, continued increases in education above some threshold may end up displacing a large enough proportion of the population into the high-income proportion, reverting the direction of inequality. In such a case, immobility keeps on rising with improvements in education, but inequality falls. Such economy is no longer *e*-Gatsbian.

In what follows, we focus, for a two-income economy, on the probability function $p(\cdot)$ and investment cost characteristics $y = \{\kappa, c_0, c_1\}$, to show the existence of transition paths when educations e changes.

Proposition 9 (Existence of Reversals)

*There exists classes of economies $(p(\cdot), y)$ with a critical level of education e^c such that the economies are *e*-Gatsbian only if $e < e^c$. Conversely, there exists classes of economies $(\tilde{p}(\cdot), \tilde{y})$ with a critical $\tilde{e}^c$ that are *e*-Gatsbian only if $e > \tilde{e}^c$.*

Proof Consider the two following probability classes:

$$p(\cdot) = (\cdot)^\sigma, \quad \sigma \in]0, 1[, \quad \tilde{p}(\cdot) = \ln((\cdot) - 1). \quad (21)$$

For each class, the relevant difference in marginal probabilities $(a+1)p^{a'} - p^{0'}$ is given by:

$$(a+1)p^{a'} - p^{0'} = [(a+1)^\sigma - 1]\sigma e^{\sigma-1} > 0, \quad (22)$$

$$(a+1)\tilde{p}^{a'} - \tilde{p}^{0'} = \frac{a+1}{(a+1)e-1} - \frac{1}{e-1} < 0. \quad (23)$$

Hence, an increase in education can only lead to an increase in social immobility I with the first class, while it can only lead to a decrease in social immobility I with the second class. However, as education e increases, the steady-state proportion Z^{2*} increases and so the sign of the derivative of inequality (dG/de) changes. Thus, in the first case, the economy is *e*-Gatsbian when $0 < Z^{2*} < \hat{Z}_1^2$, and it ceases to be so when $\hat{Z}_1^2 < Z^{2*} < 1$. The converse is true for the second case.

Because both cases yield continuous differences in marginal probabilities as a function of e , it must be that the thresholds e^c and $\tilde{e}^c$ exist. The result follows.

We now go through two examples using the probability classes of the proof of Proposition 9. The parameters specifications are presented in Table 4. $\square$

Table 4 Numerical examples

	From <i>e</i> -Gatsbian to non-Gatsbian	From non-Gatsbian to <i>e</i> -Gatsbian
$p((a+1)e)$	$\sqrt{(a+1)e}$	$\ln((a+1)e-1)$
w^2	4	4
w^1	1	1
a	3	1/2
κ	2	1
c_1	1/4	1/2
e	$e \in [0, 12 - 8\sqrt{2}]$	$e \in [2, 2.2468]$

The range for e is chosen so as to avoid corner solutions. In the second case, $e = 2.2468$ is roughly the solution to $Z^{2*} = 1$ with the chosen parameters

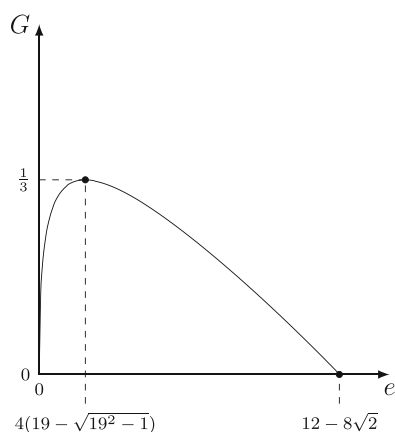
The results for $p((a+1)e) = \sqrt{(a+1)e}$ are presented in Fig. 6. Figure 6a depicts the Gini coefficient of the economy when education increases. Starting from $e = 0$, we note that the initial increases in education translate into an increase in inequality. However, when e reaches a level of $e^c = 4(19 - \sqrt{19^2 - 1})$, inequality reaches a maximum ($G = 1/3$) and then declines with further increases in education. As for Fig. 6b, it depicts immobility as a function of education. With probability function $p((a+1)e) = \sqrt{(a+1)e}$, social immobility is linearly increasing in education ($I = 1 + e/4$).

The interest here is in the combination of the two panels. In Figure 8a one can immediately note that for $e \in [0, 4(19 - \sqrt{19^2 - 1})]$, both inequality and immobility rise with education so that this segment of the e -expansion path is positively sloped and the economy is e -Gatsbian. However, when $e > 4(19 - \sqrt{19^2 - 1})$, inequality falls while immobility keeps on rising, so that portion of the e -expansion path is negatively sloped and the economy is no longer Gatsbian. When education increases steadily, as has been observed around the world in the last decades, it is therefore possible for an economy to transition from a positively-sloped segment of its e -expansion path, to a negatively-sloped one. In other words, the economy may transition from being e -Gatsbian to non-Gatsbian.

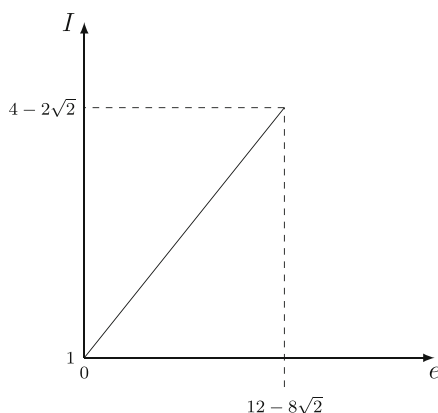
We report the results of the second example ($\tilde{p}((a+1)) = \ln((a+1)e)$) in Fig. 7. Figure 7a depicts the Gini coefficient of the economy as education increases. Starting from $e = 2$, we note that the first increases in education translate into an increase in inequality. However, when e reaches a level of $\tilde{e}^c \approx 2.043$, inequality reaches a maximum ($G = 1/3$) and then declines with further increases in education. As for the behaviour of inequality, the mechanism at work here is the same as that in Fig. 6.

However, Fig. 7b shows that for $\tilde{p}((a+1)) = \ln((a+1)e)$, immobility decreases with education. The curve depicted is a function of the difference of probabilities with e (i.e. $p^a - p^0 = \ln((3/2e - 1)/(e - 1))$).

Combining the two panels yields Fig. 8b. For $e \in [2, 2.043]$, the e -expansion path is negatively sloped, but for $e \in [2.043, 2.248]$, the path is positively sloped and the economy is e -Gatsbian. Thus, in this example, the economy transitions from being non-Gatsbian to e -Gatsbian.

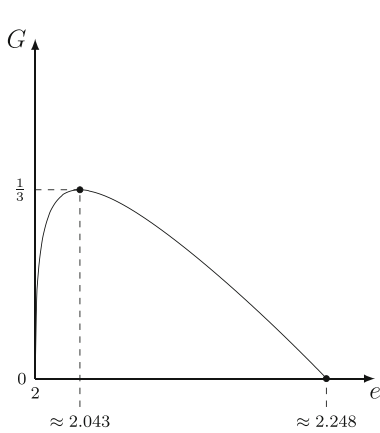
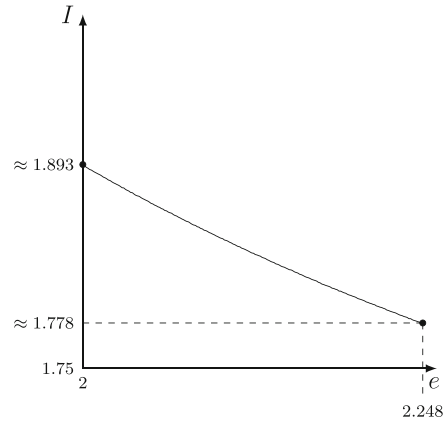


(a) Gini Coefficient as a Function of e



(b) Immobility as a Function of e

Fig. 6 $p((a+1)e) = \sqrt{(a+1)e}$

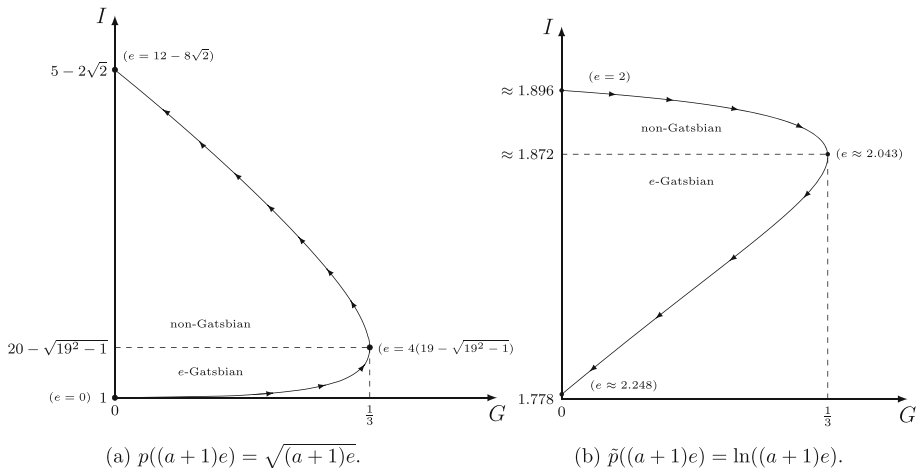
(a) Gini Coefficient as a Function of e (b) Immobility as a Function of e **Fig. 7** $\tilde{p}((a+1)) = \ln((a+1)e - 1)$

7 Conclusion

We have characterized the conditions under which Gatsby curves exist.

Our analysis builds on a rich yet tractable (Becker and Tomes 1979) environment in which parental investment and education both affect children's expected income. We are able to derive closed-form expressions for inequality and immobility. We analyze the impact of changes in the economic environment on the stationary state of the economy and we assess their impact on inequality and immobility. In the inequality-immobility space, the locus of stationary states generated by those changes is simply an expansion path. When this expansion path is positively sloped, it is a Gatsby curve.

Our results regarding the existence of Gatsby curves show the importance of accounting for both direct and indirect (composition) effects. As it turns out, the indirect effect may be

**Fig. 8** e -expansion Paths for Selected Examples

large enough to lead to counter-intuitive results. In particular, Gatsby curves may not exist as a result of the indirect effect. But for the case where the indirects effect goes in the same direction as the direct effect, as stipulated in Proposition 8, we obtain that a Gatsby curve obtains following an increase in some high income or following an increase in education. Alternatively, we show that if a Gatsby curve does not obtain following a change in the lowest income, then it must be the case that at least some composition effects on inequality go in a direction opposite to that of the direct effect.

In the empirical literature, it is well documented that education (Global Change Data Lab 2019) and the income level associated with better jobs (Acemoglu and Autor 2011; Goos et al. 2009; Barany and Siegel 2018; Bourdabat et al. 2010) have both significantly increased in the last decades in several countries. We show that Gatsby curves may have been generated by such increases. At the same time, we also show that circumstances exist in which Gatsby curves will not be generated by an increase in education or in the return to high-paying jobs, which could explain that in some countries, Gatsby curves have not been observed. Finally, we also show that an economy may transition from being Gatsbian to non-Gatsbian, and *vice versa*.

Are Gastby curves a good or a bad thing? Do they result from optimal policies? At a time where policymakers are showing keen interest in the Gatsby curve phenomenon, we must observe that these important questions have not yet received satisfactory answers in the literature. As it turns out, the framework developed in this paper naturally lends itself to a normative analysis of Gatsby curves and of education and taxation policies. It could therefore be used to provide answers to the above questions. However, prior to undertaking this task, one has to develop a proper measure of social welfare when both inequality and immobility may be affected by policies.¹⁹ These are promising avenues for future research.

Appendix

A.1 Parents' behaviour

Lemma 1 (Parents' Behaviour)

(w^1, c) -parents: $\phi^{1*} = 0 \forall c$

(w^j, c) -parents, $j > 1$: $\phi^{j*} = 0 \forall c \in [c_0, \hat{c}^j[$, and $\phi^{j*} = a \forall c \in [\hat{c}^j, c_1]$.

Proof For (w^1, c) -parents, since $\hat{c}^1 > c_1$, we have $(\kappa - c_1 w^1)a > (\kappa - \hat{c}^1 w^1)a = \mathbb{E}^\Delta$. Thus, $\mathbb{E}^\Delta - (\kappa - c_1 w^1)a < 0$ and those parents choose to invest 0. The same can be said of (w^j, c) -parents when $c \in [c_0, \hat{c}^j)$. If this is not the case, then $(\kappa - c w^j)a < (\kappa - \hat{c}^j w^j)a = \mathbb{E}^\Delta$ and those parents invest a .

A.2 Comparative statics of $\hat{c}^k$

It is useful to characterize how (w^k, c) -parents react to changes in the economic environment by analyzing the change in $\hat{c}^k$ with respect to exogenous variables. One can easily find:

$$\frac{\partial \hat{c}^k}{\partial e} = -\frac{\sum_{j=2}^n \Delta_1^j}{a w^k} [(a+1)p^{a'} - p^{0'}] \geq 0$$

¹⁹ See Gottschalk and Spolaore (2002) for an interesting analysis of the key issues and difficulties that arise in this context.

$$\begin{aligned}\frac{\partial \hat{c}^k}{\partial w^j} &= -\alpha^j \frac{p^a - p^0}{aw^k} < 0, \forall j > 1, j \neq k, \\ \frac{\partial \hat{c}^k}{\partial w^k} &= -\frac{\hat{c}^k}{w^k} - \alpha^k \frac{p^a - p^0}{aw^k} < 0, \\ \frac{\partial \hat{c}^k}{\partial w^1} &= \frac{p^a - p^0}{aw^k} > 0.\end{aligned}$$

A.3 Proof of Proposition 1

Proof Using Eq. 6, the following comparative static results can be obtained:

$$\frac{dI}{dw^1} = -\frac{(p^a - p^0)^2}{a} \sum_{i=2}^n \alpha^i \sum_{j=2}^n \frac{\alpha^j}{w^j} < 0, \quad (24)$$

$$\frac{dI}{dw^k} = (p^a - p^0) \alpha^k \left[\frac{(p^a - p^0)}{a} \sum_{j=2}^n \frac{\alpha^j}{w^j} + \frac{\hat{c}^k}{w^k} \right] > 0 \quad k \neq 1, \quad (25)$$

$$\frac{dI}{de} = \underbrace{\left[\sum_{i=2}^n \alpha^i (\bar{c} - \hat{c}^i) + \frac{(p^a - p^0)}{a} \sum_{i=2}^n \frac{\alpha^i}{w^i} \sum_{j=2}^n \alpha^j \Delta_1^j \right]}_{(+)} [(a+1)p^{a'} - p^{0'}] \geq 0. \quad (26)$$

The impacts described in the proposition follow from signing the various terms in those equations.

A.4 Proof of Proposition 2

Proof Note first that for any exogenous variable $x \in \{w^1, w^2, \dots, w^n, e\}$, we have the following relationships in the changes of the steady-state proportions:

$$\frac{\partial Z^{1*}}{\partial x} = \left(-\sum_{i=2}^n \alpha^i \right) \frac{1}{\alpha^k} \frac{\partial Z^{k*}}{\partial x}, \quad k \geq 2, \quad (27)$$

$$\frac{\partial Z^{j*}}{\partial x} = \frac{\alpha^j}{\alpha^k} \frac{\partial Z^{k*}}{\partial x}, \quad j, k \geq 2. \quad (28)$$

Turning to the comparative statics of steady-state proportions, we obtain the following:

$$\frac{\partial Z^{1*}}{\partial w^1} = -\sum_{i=2}^n \frac{Z^{i*}}{(2-I)} \frac{dI}{dw^1} > 0,$$

$$\frac{\partial Z^{1*}}{\partial w^k} = -\sum_{i=2}^n \frac{Z^{i*}}{(2-I)} \frac{dI}{dw^k} < 0, \quad k \neq 1,$$

$$\frac{\partial Z^{1*}}{\partial e} = -\sum_{i=2}^n \frac{Z^{i*}}{(2-I)} \frac{dI}{de},$$

$$\frac{\partial Z^{i*}}{\partial w^1} = \frac{Z^{i*}}{2-I} \frac{dI}{dw^1} < 0, \quad i \neq 1$$

$$\frac{\partial Z^{i*}}{\partial w^k} = \frac{Z^{i*}}{(2-I)} \frac{dI}{dw^k} > 0, \quad i, k \neq 1$$

$$\frac{\partial Z^{i*}}{\partial e} = \frac{Z^{i*}}{(2-I)} \frac{dI}{de} + \frac{\alpha^i p^{0r}}{(2-I)}, \quad i \neq 1.$$

Note that if $[(a+1)p^{a'} - p^{0r}] > 0$ (education as a complement), then $\partial Z^{1*}/\partial e < 0$ and $\partial Z^{i*}/\partial e > 0$ for $i \neq 1$. However, if $[(a+1)p^{a'} - p^{0r}] < 0$ (education as a substitute), then $\partial Z^{1*}/\partial e$ and $\partial Z^{i*}/\partial e$ for $i \neq 1$ are of ambiguous sign.

A.5 Inequality G: formula for the Gini coefficient

Denote the share of total income of group i by $\pi^i \equiv w^i Z^i / \bar{w}$. Thus, the cumulative share of total income up to group i is given by $\sum_{j=1}^i \pi^j = (1/\bar{w}) \sum_{j=1}^i w^j Z^j$. The Gini coefficient is given by $1 - 2B$, where B is the surface below the Lorenz curve. For position $i > 0$, the area of the parallelogram between i and $i-1$ is given by:

$$\frac{1}{2} \left(\sum_{j=0}^i \pi^j + \sum_{j=0}^{i-1} \pi^j \right) \left(\sum_{j=0}^i Z^j - \sum_{j=0}^{i-1} Z^j \right) = \frac{1}{2\bar{w}} \left(w^i Z^i + 2 \sum_{j=0}^{i-1} w^j Z^j \right) Z^i,$$

with the convention that $w^0 = \pi^0 = Z^0 = 0$. Hence, the Gini coefficient G is:

$$\begin{aligned} G &= 1 - 2 \left(\frac{1}{2\bar{w}} \sum_{i=1}^n \left(w^i Z^i + 2 \sum_{j=0}^{i-1} w^j Z^j \right) Z^i \right), \\ \Rightarrow G\bar{w} &= \bar{w} - \sum_{i=1}^n \left(w^i Z^i + 2 \sum_{j=0}^{i-1} w^j Z^j \right) Z^i, \\ &= \sum_{i=1}^n w^i Z^i (1 - Z^i) - 2 \sum_{i=1}^n Z^i \sum_{j=0}^{i-1} w^j Z^j, \end{aligned} \quad (29)$$

and since $Z^0 = 0$, we have:

$$G\bar{w} = \sum_{i=1}^n w^i Z^i (1 - Z^i) - 2 \sum_{i=2}^n \sum_{j=1}^{i-1} w^j Z^j Z^i. \quad (30)$$

Using $Z^1 = 1 - \sum_{i=2}^n Z^i$ and manipulating, one can obtain:

$$G\bar{w} = \sum_{i=2}^n \Delta_1^i Z^i (1 - Z^i) - 2 \sum_{i=3}^n \sum_{j=2}^{i-1} \Delta_1^j Z^j Z^i, \quad (31)$$

which is the expression used in the main text.

For the two-income case, simply note that the last part of the formula amounts to a nonsensical sum, so we obtain:

$$G\bar{w} = \Delta_1^2 Z^2 (1 - Z^2). \quad (32)$$

A.6 Inequality G: the Gini coefficient is a concave function

Consider the vectors $Z = (Z^2, \dots, Z^n)$, $Z' = (Z'^2, \dots, Z'^n)$ and their linear combination: $\alpha Z + (1 - \alpha)Z'$, $\alpha \in [0, 1]$. By linearity, we immediately get that $\bar{w}(\alpha Z + (1 - \alpha)Z') = \alpha \bar{w}(Z) + (1 - \alpha)\bar{w}(Z')$. Furthermore, recall that the right hand-side of the Gini formula is:

$$\begin{aligned} \sum_{i=2}^n \Delta_1^i Z^i (1 - Z^i) - 2 \sum_{i=3}^n \sum_{j=2}^{i-1} \Delta_1^j Z^j Z^i &= (\bar{w} - w^1) - \sum_{i=2}^n \Delta_1^i (Z^i)^2 - 2 \sum_{i=3}^n Z^i \sum_{j=2}^{i-1} \Delta_1^j Z^j, \\ &= (\bar{w} - w^1) Z^1. \end{aligned} \quad (33)$$

From this last expression, we can easily deduce that the right-hand side is a concave function:

$$\begin{aligned} & \left(\alpha(\bar{w} - w^1) + (1 - \alpha)(\bar{w}' - w'^1) \right) (\alpha Z^1 + (1 - \alpha)Z'^1) = \\ & \alpha(\bar{w} - w^1)(\alpha Z^1 + (1 - \alpha)Z'^1) + (1 - \alpha)(\bar{w}' - w'^1)(\alpha Z^1 + (1 - \alpha)Z'^1) \\ & \alpha(\bar{w} - w^1)\alpha Z^1 + \alpha(\bar{w} - w^1)(1 - \alpha)Z'^1 + (1 - \alpha)(\bar{w}' - w'^1)\alpha Z^1 + (1 - \alpha)(\bar{w}' - w'^1)(1 - \alpha)Z'^1 > \\ & \alpha(\bar{w} - w^1)\alpha Z^1 + (1 - \alpha)(\bar{w}' - w'^1)(1 - \alpha)Z'^1, \end{aligned} \quad (34)$$

which implies that the right-hand side is higher at the linear combination than at the extreme. Because $\bar{w}$ is a linear function, we can then deduce that G must be higher at the linear combination. Since the Gini coefficient represents a surface (a positive number), we can conclude that the function is non-negative and strictly concave.

A.7 Inequality G: roots of the Gini coefficient

The roots of Gini coefficient G are vectors $Z = (Z^2, \dots, Z^n)$ satisfying:

$$G = 0 \quad \Leftrightarrow \quad 0 = \sum_{i=2}^n \Delta_1^i Z^i (1 - Z^i) - 2 \sum_{i=3}^n \sum_{j=2}^{i-1} \Delta_1^j Z^j Z^i. \quad (35)$$

Note first that $\mathbf{0} \in \mathbb{R}^n$ is a trivial root. Now, consider the vector $\mathbb{1}^i \in \mathbb{R}^n$ consisting of zeros everywhere, but with a “1” in position i . Any such vector is also a root of G . Indeed, if $Z^j = 1$, then $Z^i = 0$. The converse is also true. Hence, the last term is always equal to zero. Moreover if $Z^j = 1$, then $1 - Z^j = 0$. Thus, the first summation is always zero as well.

Now, these are the sole roots of equation (35). To see this, note that equation (35) is a quadratic form of $n - 1$ variables. This means that it has at most $2(n - 1)$ roots (2 per variable), but since $\mathbf{0}$ is a root for each of the $n - 1$ variables, at most $n - 1$ roots remain. Since we have already found those remaining roots ($\mathbb{1}^i$, $i = 2 \dots n$), it follows that no other roots exist.

A.8 Inequality G: partial derivatives with respect to proportions

The partial derivatives of G with respect to Z^i are as follows:

$$\begin{aligned} \frac{\partial G}{\partial Z^2} + G \Delta_1^2 &= \Delta_1^2 (1 - 2Z^2) - 2 \sum_{i=3}^n \Delta_1^2 Z^i, \\ \Rightarrow \frac{\partial G}{\partial Z^2} &= \Delta_1^2 \left(1 - G - 2 \sum_{i=2}^n Z^i \right), \end{aligned} \quad (36)$$

$$\begin{aligned}\frac{\partial G}{\partial Z^k} + G\Delta_1^k &= \Delta_1^k(1 - 2Z^k) - 2\left(\sum_{j=2}^{k-1} \Delta_1^j Z^j + \sum_{j=k+1}^n \Delta_1^k Z^j\right), \\ \Rightarrow \frac{\partial G}{\partial Z^k} &= \Delta_1^k \left(1 - G - 2\sum_{j=k}^n Z^j - 2\sum_{j=2}^{k-1} \frac{\Delta_1^j}{\Delta_1^n} Z^j\right),\end{aligned}\quad (37)$$

$$\begin{aligned}\frac{\partial G}{\partial Z^n} + G\Delta_1^n &= \Delta_1^n(1 - 2Z^n) - 2\sum_{j=2}^{n-1} \Delta_1^j Z^j, \\ \Rightarrow \frac{\partial G}{\partial Z^n} &= \Delta_1^n \left(1 - G - 2Z^n - 2\sum_{j=2}^{n-1} \frac{\Delta_1^j}{\Delta_1^n} Z^j\right), \\ &= \Delta_1^n \left(1 - G - 2Z^n - 2\sum_{j=3}^n \frac{\Delta_1^{j-1}}{\Delta_1^n} Z^{j-1}\right).\end{aligned}\quad (38)$$

Summing up the equations, we obtain:

$$\begin{aligned}\frac{\partial G}{\partial Z^k} &= \Delta_1^k \left(1 - G - 2\sum_{j=k}^n Z^j - 2\sum_{j=2}^{k-1} \frac{\Delta_1^j}{\Delta_1^n} Z^j\right), \\ &= \Delta_1^k \left(\sum_{j=1}^{k-1} Z^j - \sum_{j=k}^n Z^j - 2\sum_{j=2}^{k-1} \frac{\Delta_1^j}{\Delta_1^n} Z^j - G\right).\end{aligned}\quad (39)$$

where we ignore the summations with inverted indices (at $j = 2$ or $j = n$).

Proof of Proposition 3. Subtracting two partial derivatives, we obtain:

$$\begin{aligned}\frac{\partial G}{\partial Z^{k+1}} - \frac{\partial G}{\partial Z^k} &= \Delta_1^{k+1} \left(1 - G - 2\sum_{j=k+1}^n Z^j - 2\sum_{j=2}^k \frac{\Delta_1^j}{\Delta_1^n} Z^j\right) \\ &\quad - \Delta_1^k \left(1 - G - 2\sum_{j=k}^n Z^j - 2\sum_{j=2}^{k-1} \frac{\Delta_1^j}{\Delta_1^n} Z^j\right), \\ &> \Delta_1^{k+1} \left(1 - G - 2\sum_{j=k+1}^n Z^j - 2\sum_{j=2}^k \frac{\Delta_1^j}{\Delta_1^n} Z^j\right) \\ &\quad - \Delta_1^{k+1} \left(1 - G - 2\sum_{j=k}^n Z^j - 2\sum_{j=2}^{k-1} \frac{\Delta_1^j}{\Delta_1^n} Z^j\right), \\ &= 2\Delta_1^{k+1} \frac{\Delta_k^n}{\Delta_1^n} Z^k > 0.\end{aligned}\quad (40)$$

Thus, the derivatives are ordered: $\partial G/\partial w^j > \partial G/\partial w^i$ if $j > i$. The sole exception is when $Z^k = 0$, for all $k < n$, as the partial derivatives then boil down to:

$$0 = 1 - G - 2Z^n, \quad (41)$$

We will see below that this point is the global maximum of G .

A.9 Inequality G: maxima

The partial derivatives of the Gini coefficient are ordered, in the sense that the set of equations has no general interior solution. However, at the point $Z^n = (1 - G)/2$ (and all other coordinates at zero), all the equations solve to zero, implying that this point is the global maximum.

Solving yields:

$$\bar{w}(1 - 2Z^n) = G\bar{w}, \quad \Rightarrow w^1 - 2w^1Z^n - \Delta_1^n(Z^n)^2 = 0,$$

which sole positive root is:

$$\hat{Z}_1^n = \frac{2w^1 - \sqrt{4(w^1)^2 + 4w^1\Delta_1^n}}{-2\Delta_1^n} = \frac{\sqrt{w^1}}{\sqrt{w^n} + \sqrt{w^1}} < \frac{1}{2}. \quad (42)$$

By substitution, one can find that the Gini coefficient satisfies:

$$G_1^n = \frac{\sqrt{w^n} - \sqrt{w^1}}{\sqrt{w^n} + \sqrt{w^1}}. \quad (43)$$

Note that all the other partial derivatives can be solved independently by setting $Z^k = (1 - G)/2$, $k \neq n$. In such a case, only one partial derivative is different from zero, and it yields:

$$\hat{Z}_1^k = \frac{\sqrt{w^1}}{\sqrt{w^1} + \sqrt{w^k}}, \quad G_1^k = \frac{\sqrt{w^k} - \sqrt{w^1}}{\sqrt{w^k} + \sqrt{w^1}}. \quad (44)$$

Note that $G_1^n > G_1^{n-1} > \dots > G_1^2$.

Those are the sole maxima that can be found with reference to w^1 . However, by using a transformation such as $Z^j = 1 - \sum_{i=1, i \neq j}^n Z^i$, a simple exploitation of the symmetry of the Gini formula allows us to find other maxima:

$$\hat{Z}_j^i = \frac{\sqrt{w^j}}{\sqrt{w^j} + \sqrt{w^i}}, \quad G_j^i = \frac{\sqrt{w^i} - \sqrt{w^j}}{\sqrt{w^i} + \sqrt{w^j}} \quad \forall j < i. \quad (45)$$

Since there are $n(n - 1)/2$ possible combinations of indices $2 < i < n$, $1 < j < i$, the total number of maxima that can be found is $n(n - 1)/2$.

A.10 Inequality G: partial derivatives with respect to incomes

Proof of Proposition 4 We can find $\partial G / \partial w^k$ by differentiating Eq. 29, which, for $i < n$, leads to Eq. 14 in the main text. It can be easily verified that this expression is negative for $i = 1$. The specific expression for $i = n$ is:

$$\frac{\partial G}{\partial w^n} \bar{w} = Z^n(1 - G - Z^n). \quad (46)$$

It is positive if $1 - Z^n > G$, which is equivalent to:

$$\begin{aligned} (1 - Z^n) &> G, \\ \Leftrightarrow \bar{w}(1 - Z^n) &> G\bar{w}, \end{aligned}$$

$$\begin{aligned}
&\Leftrightarrow w^1(1 - Z^n) + \sum_{i=2}^n \Delta_1^i Z^i (1 - Z^n) > \sum_{i=2}^n \Delta_1^i Z^i (1 - Z^i) - 2 \sum_{i=3}^n \sum_{j=2}^{i-1} \Delta_1^j Z^j Z^i, \\
&\Leftrightarrow w^1(1 - Z^n) > - \sum_{j=2}^{n-1} \Delta_1^j Z^j Z^n - \sum_{i=2}^{n-1} \Delta_1^i (Z^i)^2 - 2 \sum_{i=3}^{n-1} \sum_{j=2}^{i-1} \Delta_1^j Z^j Z^i < 0, \quad (47)
\end{aligned}$$

which is always true.

A.11 Proof of Proposition 6

Proof We can combine (16) with (28) to obtain:

$$\frac{dG}{dx} \bar{w} = \frac{1}{\alpha^2} \frac{\partial Z^{2*}}{\partial x} \sum_{i=2}^n \alpha^i \Delta_1^i \left[1 - G - 2 \sum_{j=i}^n Z^j - 2 \sum_{j=2}^{i-1} \frac{\Delta_1^j}{\Delta_1^i} Z^j \right] + \frac{\partial G}{\partial x}. \quad (48)$$

Then note that if the economy is not w^k -dominated for all $k > 1$, then the indirect effect amounts to the composition effects pre-multiplied by a positive number. Hence, the change in inequality inherits the sign of the change in proportions (which, in the case of incomes, is the same as the direct effect on inequality). Conversely, if the change in inequality is negative, then the indirect effect must be negative, which means that there is at least one income $k > 1$ such that the economy is w^k -dominated.

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